

$\underline{\sigma} \rightarrow$ tensor de tensiones

$\frac{\partial u_i}{\partial x_j \partial x_k}$

$(\underline{u} \times \underline{v})_i = \epsilon_{ijk} \cdot u_j v_k$

a. $u_i = e_{ijk} \tau_{jk} = \frac{1}{2} (\epsilon_{ijk} \sigma_{jk} + \epsilon_{ijk} \sigma_{jk}) = \frac{1}{2} (\epsilon_{ijk} \sigma_{jk} - \epsilon_{ikj} \sigma_{kj}) = 0 \Rightarrow \underline{u} = \underline{0}$

σ es simétrico $\Rightarrow \sigma_{jk} = \sigma_{kj}$

b. $e_{ijk} e_{ist} \tau_{kt} =$

$u_{js} = e_{ijk} e_{ist} \sigma_{kt} = (\sigma_{js} \sigma_{kt} - \sigma_{jt} \sigma_{ks}) \sigma_{kt} = \sigma_{js} \cdot \sigma_{tt} - \sigma_{js}$

$= \sigma_{js} \cdot \sigma_{tt} - \sigma_{js}$

3 tensión media (σ_m)

traza $\rightarrow$ suma de los
elementos de la
diagonal

$\underline{D} = \underline{\sigma} - \frac{1}{3} \sigma_m \underline{I}$

$D_{js} = \sigma_{js} - \frac{1}{3} \sigma_{tt} \cdot \delta_{js}$